SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I Juhi Surin (192525148) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Juhi Surin

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10

5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a welldesigned conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.

Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

Question 1
Problem Analysis and Agent Design
AI-Based Student Support Assistant

1. Problem Analysis

Every semester, universities deal with an overwhelming number of student queries — course registration status, class timetables, exam schedules, attendance percentages, fee dues, hostel allocation, and general campus services. Individually, none of these questions are hard to answer. The real problem is volume: the same handful of questions get repeated by hundreds of students within the same few days, usually right around registration week or just before exams.

In the current manual setup, a student has to visit the admin office in person, send an email and wait, or track down a faculty coordinator — all of which only work within office hours. This creates several concrete limitations:

- Limited office working hours — no support at night, on weekends, or during holidays.
- Slow response to urgent queries — a student who needs today's exam hall number can't afford to wait a day for a reply.
- Human errors — a tired or overloaded staff member can quote outdated attendance figures or the wrong exam date.
- Heavy workload on administrative staff — the same repetitive questions eat into time that could go toward genuinely complex cases.
- Reduced student satisfaction — long queues and delayed replies leave students frustrated during an already stressful period.

Figure 1 lays out what this manual chain actually looks like end to end, and why it slows down so badly under load.

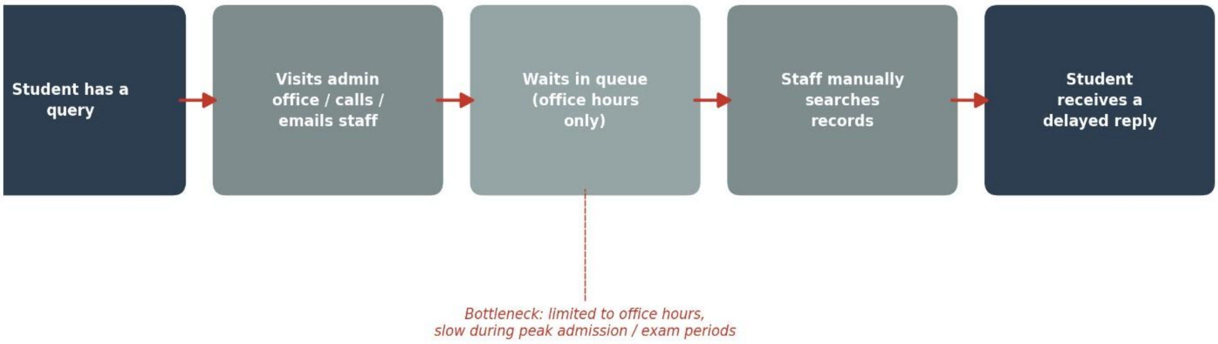

Figure 1: Workflow of the existing manual student support system

To fix this, the university can implement an AI-Based Student Support Assistant using Dialogflow. Instead of matching rigid keywords, the chatbot applies Artificial Intelligence and Natural Language Processing to understand what a student is actually asking, work out the intent behind it, pull out the relevant details, and reply instantly — available 24×7, to as many students as need it at once, with consistent and accurate answers every time.

2. How Artificial Intelligence Solves the Problem

What makes AI the right fit here — rather than just a nice-to-have — is that it lets the chatbot hold something closer to a real conversation instead of relying on exact keyword matches. A student can phrase the same question a dozen different ways ("what's my attendance," "am I short on attendance," "how many classes have I missed"), and the agent still has to land on the same intent. Specifically, the AI layer is doing the following work:

- Understands free-form messages using Natural Language Processing (NLP), instead of requiring an exact phrase.
- Performs Intent Recognition to figure out what the student actually wants — registration, timetable, attendance, and so on.
- Uses Entity Extraction to pull out specific details from the sentence, like Student ID, Course Code, Semester, or Department.
- Retrieves accurate, current information directly from the university database rather than relying on memory.
- Learns from additional training phrases over time, so it gets better at handling new ways of asking the same question.
- Runs continuously without needing a human on the other end, which is what makes 24×7 support realistic in the first place.

Put together, this is what actually solves the three core problems from Section 1: delay disappears because replies take seconds, staff overload eases because the bot silently absorbs the repetitive queries, and inconsistency drops because every student's answer comes from the same live database rather than a person's recollection. Figure 2 shows how this proposed system replaces the slow manual chain from Figure 1.

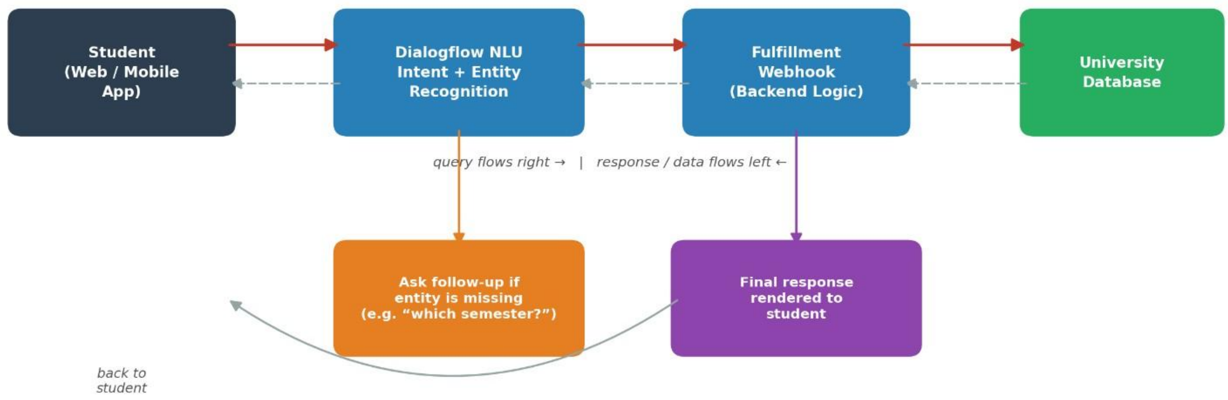

Figure 2: Proposed AI-based student support assistant architecture

3. Agent Design

Rather than describing the chatbot loosely as "a bot that answers questions," it's more useful — and more accurate, in AI terms — to design it as a rational agent with a clearly defined objective, user base, environment, and set of inputs and outputs. That's what actually drives implementation decisions like which platform to use and which intents to build first.

Objective

To design an intelligent conversational agent that provides instant, accurate, and personalised responses to student queries regarding course registration, timetable, examination schedule, attendance, fee details, hostel services, library services, and other campus facilities — while reducing administrative workload and offering round-the-clock support.

Users

The system is built primarily for students, but it's used by a wider group in practice:

- Students — the main users, asking about their own academic and administrative records.
- Faculty Members — checking timetable slots or attendance data for their sections.
- Administrative Staff — receiving queries the bot escalates rather than answers itself.
- Examination Cell, Admission Office, and Library Staff — each handling escalations specific to their department.

Each of these groups interacts with the same chatbot but triggers different intents depending on what they need, which is why the intent list has to be organised by department rather than treated as one flat list.

Environment

The agent operates inside the university's digital ecosystem — accessible through the university website, the student portal, and a mobile application, all connected over the internet. Behind the scenes, it talks to the Dialogflow platform and the university database through webhooks to fetch real, current data rather than pre-written answers.

In AI terms, this environment is partially observable — the agent only knows what the current conversation and the database expose, not the student's full history — and stochastic, since the same request can arrive phrased in unpredictable ways. It's also effectively multi-agent, since many students hold independent conversations with the system at the same time. This is precisely why a fixed, rule-based script would fall short here: the environment demands a model that can handle uncertainty and varied phrasing, which is what justifies using Dialogflow's NLU engine instead of a simple keyword lookup.

Inputs

The chatbot accepts natural-language queries from users, along with structured details it extracts from that text, such as Student ID, Department, Semester, Course Code, Examination Name, Attendance Request, Fee Query, Library Request, or Campus Service Request. In practice, a student's input might look like:

- "What is my attendance percentage?"
- "When is my AI examination?"
- "Register my Semester 5 courses."
- "Show today's timetable."

Outputs

After processing a request, the chatbot returns a specific, meaningful response rather than a generic acknowledgement — course registration confirmation, class timetable, examination schedule, attendance percentage, fee payment status, library book availability, hostel allocation details, scholarship status, campus event notifications, or department contact information. If the requested information genuinely isn't available to it, the chatbot doesn't guess — it politely redirects the student to the right department instead.

4. Expected Performance Measures

Judging the agent only by whether it "works" isn't good enough — its effectiveness needs to be measured against specific, checkable targets. These are the measures that matter most for a system like this:

Performance Measure	Why It Matters
Response Accuracy	Correctly identifies student intent and returns the right information, minimising misunderstandings.
Response Time	Generates a reply within a few seconds of the query being sent, so students never feel like they're waiting on a person.
User Satisfaction	Answers are relevant and easy to read, not just technically correct — measured through follow-up feedback prompts.

Availability	Runs 24×7, including weekends, holidays, and late nights when the admin office is closed.
Scalability	Handles spikes in traffic during registration week or result day without slowing down.
Reliability	Always pulls from the live university database, so information never goes stale the way a human's memory might.
Maintainability	New intents and training phrases can be added quickly whenever policies, courses, or schedules change.
Security	Student-specific data (attendance, fees) is only released after authentication, over encrypted channels.

5. PEAS Representation

Putting the objective, environment, inputs, and outputs together in one place gives the standard PEAS description of the agent, shown below alongside its visual form in Figure 3.

PEAS Component	Description
Performance Measure	High response accuracy, fast reply time, user satisfaction, continuous availability, reliability, and data security.
Environment	University website, student portal, mobile application, the Dialogflow platform, the university database, and the internet connecting them.
Actuators	Chatbot text replies, push notifications, confirmation messages, hyperlinks to forms, and downloadable documents.
Sensors	Student text queries, student ID, course code, semester, department, menu selections, and responses returned from the database.

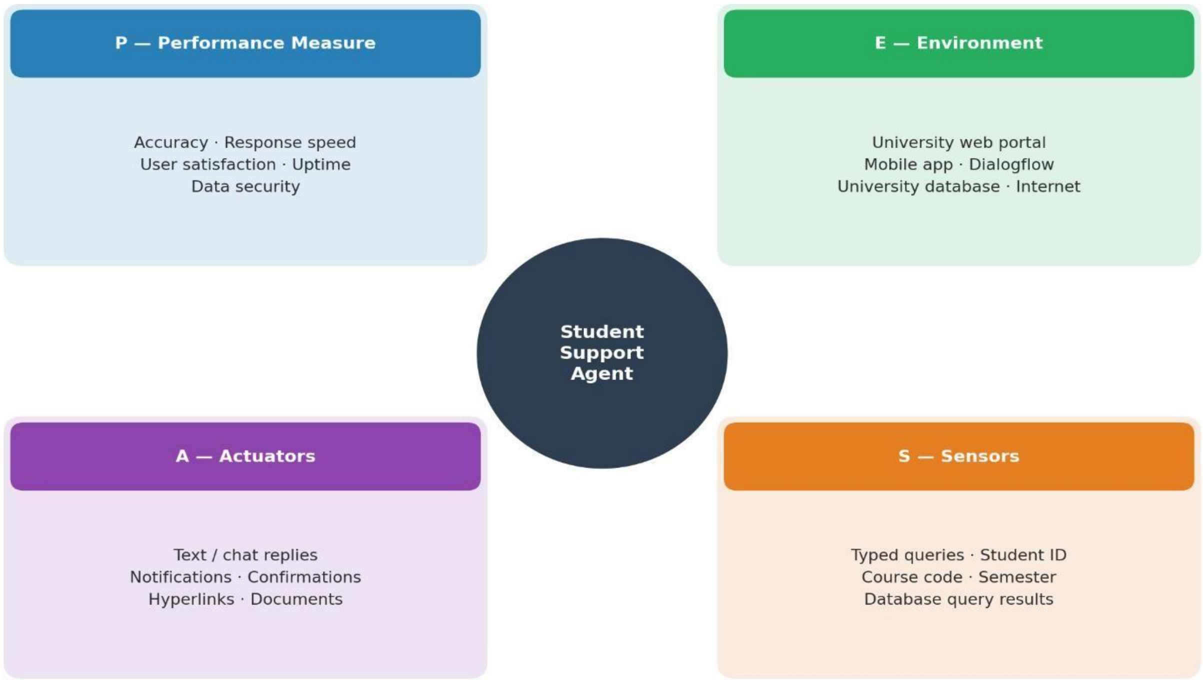

Figure 3: PEAS model of the AI-based student support assistant

6. Advantages of the Proposed AI Agent

- Provides instant responses to student queries instead of making them wait for staff availability.
- Reduces administrative workload by silently absorbing repetitive, low-complexity questions.
  - Available 24×7 without interruption, including nights, weekends, and holidays.
- Improves communication between students and the university through a single, consistent channel.
- Reduces human error by pulling answers from authorised, live data instead of memory.
- Supports multiple students simultaneously without any drop in response quality.
- Delivers consistent, personalised responses regardless of who — or what time — is asking.
- Frees up staff time for the complex, judgement-based cases that genuinely need a human.

7. Conclusion

The manual student support process doesn't fail because staff are careless — it fails because it forces a high volume of simple, repetitive queries through a channel that only a limited number of humans, working limited hours, can serve. Designing the fix as a properly defined AI agent — with a clear objective, users, environment, inputs, outputs, and performance measures, expressed through the PEAS model — makes every implementation choice traceable back to a reason: NLU because the environment is stochastic and free-form, webhook-based retrieval because the environment is only partially observable to the agent on its own, and department-wise escalation because the performance measure includes reliability, not just speed. Built this way, the AI-Based Student Support Assistant gives students faster, more accurate answers around the clock while giving university staff their time back for the cases that actually need a person.

References

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